

RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL

New Scheme Based On AICTE Flexible Curricula

CSE-Artificial Intelligence/ Artificial Intelligence, VI-Semester

AI 601 Theory of Computation

Course Outcomes: After completion of the course students will be able to

- CO1 explain the basic concepts of switching and finite automata theory & languages.
- CO2 relate practical problems to languages, automata, computability and complexity.
- CO3 construct abstract models of computing and check their power to recognize the languages.
- CO4 analyze the grammar, its types, simplification and normal form.
- CO5 interpret rigorously formal mathematical methods to prove properties of languages, grammars and automata.
- CO6 develop an overview of how automata theory, languages and computation are applicable in engineering application.

Detailed Contents

Unit-I

Introduction of Automata Theory: Examples of automata machines, Finite Automata as a language acceptor and translator, Moore machines and mealy machines, composite machine, Conversion from Mealy to Moore and vice versa

Unit-II

Types of Finite Automata: Non Deterministic Finite Automata (NFA), Deterministic finite automata machines, conversion of NFA to DFA, minimization of automata machines, regular expression, Arden's theorem. Meaning of union, intersection, concatenation and closure, 2 way DFA.

Unit-III

Grammars: Types of grammar, context sensitive grammar, and context free grammar, regular grammar. Derivation trees, ambiguity in grammar, simplification of context free grammar, conversion of grammar to automata machine and vice versa, Chomsky hierarchy of grammar, killing null and unit productions. Chomsky normal form and Greibach normal form.

Unit-IV

Push down Automata: example of PDA, deterministic and non-deterministic PDA, conversion of PDA into context free grammar and vice versa, CFG equivalent to PDA, Petrinet model.

Unit-V

Turing Machine: Techniques for construction. Universal Turing machine, Multi tape, multi head and multidimensional Turing machine, N-P complete problems. Decidability and Recursively Enumerable Languages, decidability, decidable languages, undecidable languages, Halting problem of Turing machine & the post correspondence problem.

RECOMMENDED BOOKS

- Introduction to Automata Theory Language & Computation, Hopcroft & Ullman, Narosa Publication.
- Element of the Theory Computation, Lewis & Christors, Pearson.
- Theory of Computation, Chandrasekhar & Mishra, PHI.
- Theory of Computation, Wood, Harper & Row.
- Introduction to Computing Theory, Daniel I-A Cohen, Wiley.

LIST OF EXPERIMENTS

1. Design a Program for creating machine that accepts three consecutive ones.
2. Design a Program for creating machine that accepts the string always ending with 101.
3. Design a Program for Mode 3 Machine
4. Design a program for accepting decimal number divisible by 2.

5. Design a program for creating a machine which accepts a string having an equal number of 1's and 0's.
6. Design a program for creating a machine which counts the number of 1's and 0's in a given string.
7. Design a program to find 2's complement of a given binary number.
8. Design a program which will increment the given binary number by 1.
9. Design a program to convert NFA to DFA.
10. Design a program to create a PDA machine that accepts the well-formed parentheses.
11. Design a PDA to accept $WCWR$ where w is any string and WR is the reverse of that string and C is a special symbol.
12. Design a Turing machine that accepts the following language $a^n b^n c^n$ where $n > 0$

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AI 602 Computer Networks

Course Outcomes: After completion of the course students will be able to

1. Characterize computer networks from the view point of components and services.
2. Demonstrate the simulation of data flow in network protocols.
3. Select networking protocol as per the requirements.
4. Model a problem or situation in terms of layering concept and map it to the TCI/IP stack
5. Select the most suitable Application Layer protocol as per the requirements of the network application

COURSE CONTENTS:

Unit –I

Computer Network: Definitions, goals, components, Architecture, Classifications & Types, Layered Architecture: Protocol hierarchy, Design Issues, Interfaces and Services, Connection Oriented & Connectionless Services, Service primitives, Design issues & its functionality. ISOOSI Reference Model: Principle, Model, Descriptions of various layers and its comparison with TCP/IP. Principals of physical layer: Media, Bandwidth, Data rate and Modulations

Unit-II

Data Link Layer: Need, Services Provided, Framing, Flow Control, Error control. Data Link Layer Protocol: Elementary & Sliding Window protocol: 1-bit, Go-Back-N, Selective Repeat, Hybrid ARQ. Protocol verification: Finite State Machine Models & Petri net models. ARP/RARP/GARP

UNIT III:

MAC Sub layer: MAC Addressing, Binary Exponential Back-off (BEB) Algorithm, Distributed Random Access Schemes/Contention Schemes: for Data Services (ALOHA and Slotted ALOHA), for Local-Area Networks (CSMA, CSMA/CD, CSMA/CA), Collision Free Protocols: Basic Bit Map, BRAP, Binary Count Down, MLMA Limited Contention Protocols: Adaptive Tree Walk, Performance Measuring Metrics. IEEE Standards 802 series & their variant.

UNIT IV:

Network Layer: Need, Services Provided, Design issues, Routing algorithms: Least Cost Routing algorithm, Dijkstra's algorithm, Bellman-ford algorithm, Hierarchical Routing, Broadcast Routing, Multicast Routing. IP Addresses, Header format, Packet forwarding, Fragmentation and reassembly, ICMP, Comparative study of IPv4 & IPv6

UNIT V:

Transport Layer: Design Issues, UDP: Header Format, Per-Segment Checksum, Carrying Unicast/Multicast Real-Time Traffic, TCP: Connection Management, Reliability of Data Transfers, TCP Flow Control, TCP Congestion Control, TCP Header Format, TCP Timer Management. Application Layer: WWW and HTTP, FTP, SSH, Email (SMTP, MIME, IMAP), DNS, Network Management (SNMP).

References:

1. Andrew S. Tanenbaum, David J. Wetherall, "Computer Networks" Pearson Education.
2. Douglas E Comer, "Internetworking With Tcp/Ip Principles, Protocols, And Architecture - Volume I" 6

th Edition, Pearson Education

3. Dimitri Bertsekas, Robert Gallager, "Data Networks", PHI Publication, Second Edition.

4. Kaveh Pahlavan, Prashant Krishnamurthy, "Networking Fundamentals", Wiley Publication.

5. Uysal Black, "Computer Networks", PHI Publication, Second Edition.

6. Ying-Dar Lin, Ren-Hung Hwang, Fred Baker, "Computer Networks: An Open Source Approach", McGraw Hill.

List of Experiments:

1. Study of Different Type of LAN & Network Equipments.
2. Study and Verification of standard Network topologies i.e. Star, Bus, Ring etc.
3. LAN installations and Configurations.
4. Write a program to implement various types of error correcting techniques.
5. Write a program to Implement various types of framing methods.
6. Study of Tool Command Language (TCL).
7. Study and Installation of Standard Network Simulator: N.S-2, N.S3, OpNet, QualNet etc .
8. Study & Installation of ONE (Opportunistic Network Environment) Simulator for High Mobility Networks .
9. Configure 802.11 WLAN.
10. Implement & Simulate various types of routing algorithm.
11. Study & Simulation of MAC Protocols like Aloha, CSMA, CSMA/CD and CSMA/CA using Standard Network Simulators.
12. Study of Application layer protocols-DNS, HTTP, HTTPS, FTP and TelNet.

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Departmental Elective AI 603(A) Deep Learning

Course outcomes: After completing the course student will be able to:

1. Describe in-depth about theories, fundamentals, and techniques in Deep learning.
2. Identify the on-going research in computer vision and multimedia field.
3. Evaluate various deep networks using performance parameters.
4. Design and validate deep neural networks as per requirements.

Detailed Contents:

Unit I: Introduction History of Deep Learning, McCulloch Pitts Neuron, Multilayer Perceptions (MLPs), Representation Power of MLPs, Sigmoid Neurons, Feed Forward Neural Networks, Back propagation, weight initialization methods, Batch Normalization, Representation Learning, GPU implementation, Decomposition – PCA and SVD.

Unit II: Deep Feed forward Neural Networks, Gradient Descent (GD), Momentum Based GD, Nesterov Accelerated GD, Stochastic GD, AdaGrad, Adam, RMSProp, Auto-encoder, Regularization in auto-encoders, Denoising auto-encoders, Sparse auto-encoders, Contractive auto-encoders, Variational auto-encoder, Auto-encoders relationship with PCA and SVD, Dataset augmentation. Denoising auto encoders,

Unit III: Introduction to Convolutional neural Networks (CNN) and its architectures, CNN terminologies: ReLu activation function, Stride, padding, pooling, convolutions operations, Convolutional kernels, types of layers: Convolutional, pooling, fully connected, Visualizing CNN, CNN examples: LeNet, AlexNet, ZF-Net, VGGNet, GoogLeNet, ResNet, RCNN etc. Deep Dream, Deep Art. Regularization: Dropout, drop Connect, unit pruning, stochastic pooling, artificial data, injecting noise in input, early stopping, Limit Number of parameters, Weight decay etc.

Unit IV: Introduction to Deep Recurrent Neural Networks and its architectures, Backpropagation Through Time (BPTT), Vanishing and Exploding Gradients, Truncated BPTT, Gated Recurrent Units (GRUs), Long Short Term Memory (LSTM), Solving the vanishing gradient problem with LSTMs, Encoding and decoding in RNN network, Attention Mechanism, Attention over images, Hierarchical Attention, Directed Graphical Models. Applications of Deep RNN in Image Processing, Natural Language Processing, Speech recognition, Video Analytics.

Unit V: Introduction to Deep Generative Models, Restricted Boltzmann Machines (RBMs), Gibbs Sampling for training RBMs, Deep belief networks, Markov Networks, Markov Chains, Auto-regressive Models: NADE, MADE, PixelRNN, Generative Adversarial Networks (GANs), Applications of Deep Learning in Object detection, speech/ image recognition, video analysis, NLP, medical science etc.

TEXTBOOKS RECOMMENDED:

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville; Deep Learning, MIT Press.
2. Charu C. Aggarwal "Neural Networks and Deep Learning: A Textbook", Springer.
3. Francois Chollet, "Deep Learning with Python", Manning Publications.

REFERENCE BOOKS:

1. Aurelien Geon, "Hands-On Machine Learning with Scikit Learn and Tensor flow: Concepts, Tools, and Techniques to Build Intelligent Systems", O'Reilly.
2. Andreas Muller, "Introduction to Machine Learning with Python: A Guide for Data Scientists", O'Reilly.
3. Adam Gibson, Josh Patterson, "Deep Learning: A Practitioner's Approach", O'Reilly.

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CSE-Artificial Intelligence/ Artificial Intelligence, VI-Semester

Departmental Elective AI 603(B) Data and Visual Analytics

Detailed Content

Unit 1 : Data Definitions and Analysis Techniques: Elements, Variables, and Data categorization Levels of Measurement Data management and indexing Introduction to Statistical Concepts: Sampling Distributions, Resampling, Statistical Inference and Descriptive Statistics, Measures of central tendency, Measures of location of dispersions

Unit2: Advance Data analysis techniques: Statistical hypothesis generation and testing, Chi-Square test, t-Test, Analysis of variance, Correlation analysis, Maximum likelihood test, Regression Modelling, Multivariate Analysis, Bayesian Modelling, Inference and Bayesian Network, Regression analysis

Unit3: Data Wrangling: Intro to Data Wrangling, Gathering Data, Assessing Data, Cleaning Data.

Data Visualization in Data Analysis: Design of Visualizations, Univariate Exploration of Data, Bivariate Exploration of Data, Multivariate Exploration of Data, Explanatory Visualizations.

Unit 4: Data Ecosystem: Overview of the Data Analyst Ecosystem, Types of Data, Understanding Different Types of File Formats, Sources of Data, Overview of Data Repositories, NoSQL, Data Marts, Data Lakes, ETL, and Data Pipelines, Foundations of Big Data, Big Data processing tools such as Hadoop, Hadoop Distributed File System (HDFS), Hive, and Spark

Unit5: Data Visualization tools: Python visualization libraries (mat plot lib, pandas, seaborn, gg plot, plotly), Introduction to Power BI tools, Examples of inspiring (industry) projects- Exercise: create your own visualization of a complex dataset.

Text Books/References:

1. Joel Grus, Data Science from Scratch, Shroff Publisher Publisher/O'Reilly Publisher Media
2. Annalyn Ng, Kenneth Soo, Numsense, Data Science for the Layman, Shroff Publisher
3. Cathy O'Neil and Rachel Schutt. Doing Data Science, Straight Talk from The Frontline. O'Reilly Publisher Media.
4. Jure Leskovek, Anand Rajaraman and Jeffrey Ullman. Mining of Massive Datasets. v2.1, Cambridge University Press.
5. Jake Vander Plas, Python Data Science Handbook, Shroff Publisher Publisher/O'Reilly Publisher Media
6. Philipp Janert, Data Analysis with Open Source Tools, Shroff Publisher Publisher/O'Reilly Publisher Media.

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CSE-Artificial Intelligence/ Artificial Intelligence, VI-Semester

Departmental Elective AI 603(C) Pattern Recognition

Course Objective: To help students understand basic mathematical and statistical techniques commonly used in pattern recognition. To introduce students to a variety of pattern recognition algorithms.

Detailed Contents:

Module 1: Introduction and mathematical Preliminaries Principles of pattern recognition: Uses, mathematics, Classification and Bayesian rules, Clustering vs classification, Basics of linear algebra and vector spaces, Eigenvalues and eigenvectors, Rank of matrix and SVD

Module 2: Pattern Recognition basics Bayesian decision theory, Classifiers, Discriminant functions, Decision surfaces, Parameter estimation methods, Hidden Markov models, dimension reduction methods, Fisher discriminant analysis, Principal component analysis, non-parametric techniques for density estimation, non-metric methods for pattern classification, unsupervised learning, algorithms for clustering: K means, Hierarchical and other methods

Module 3: Feature Selection and extraction Problem statement and uses, Branch and bound algorithm, Sequential forward and backward selection, Cauchy Schwartz inequality, Feature selection criteria function: Probabilistic separability based and Interclass distance based, Feature Extraction: principles

Module 4: Visual Recognition Human visual recognition system, Recognition methods: Low-level modelling (e.g. features), Mid-level abstraction (e.g. segmentation), High-level reasoning (e.g. scene understanding); Detection/Segmentation methods; Context and scenes, Importance and saliency, Large-scale search and recognition, Egocentric vision, systems, Human-in-the-loop interactive systems, 3D scene understanding.

Module 5: Recent advancements in Pattern Recognition Comparison between performance of classifiers, Basics of statistics, covariance and their properties, Data condensation, feature clustering, Data visualization, Probability density estimation, Visualization and Aggregation, FCM and soft-computing techniques, Examples of real-life datasets.

Laboratory/Practicals:

1. Data extraction
2. Pre-processing of images
3. Image segmentation
4. Image classification AICTE

Text Books/Suggested References:

1. Pattern Recognition and Machine Learning by Christopher M. Bishop, Springer, 2006.
2. Pattern Classification by Richard O. Duda, Peter E. Hart, David G. Stork, Wiley, 1973.
3. <https://nptel.ac.in/courses/106/106/106106046/>

Course Outcomes: After completion of course, students would be able to:

1. Understand basic mathematical and statistical techniques commonly used in pattern recognition.
2. Apply a variety of pattern recognition algorithms.
3. Understand and apply various pre-processing algorithms.
4. Apply various algorithms for image classification.

CourseObjective: The students will be able to understand and analyze how to deal with changing data. They will also be able to identify and interpret potential unintended effects in your project. They will understand and define procedures to operationalize and maintain your applied machine learning model.

Detailed Contents:

Unit I: Introduction What is optimization, Formulation of LPP, Solution of LPP: Simplex method, Basic Calculus for optimization: Limits and multivariate functions, Derivatives and linear approximations: Single variate functions and multivariate functions.

Unit II: Machine Learning Strategy ML readiness, Risk mitigation, Experimental mindset, Build/buy/partner, setting up a team, Understanding and communicating change.

Unit III: Responsible Machine Learning AI for good and all, Positive feedback loops and negative feedback loops, Metric design and observing behaviors, Secondary effects of optimization, Regulatory concerns.

Unit IV: Machine Learning in production and planning Integrating info systems, users break things, time and space complexity in production, when to retain the model? Logging ML model versioning, Knowledge transfer, Reporting performance to stakeholders.

Unit V: Care and feeding of your machine learning model MLPL Recap, Post deployment challenges, QUAM monitoring and logging, QUAM Testing, QUAM maintenance, QUAM updating, Separating Data stack from Production, Dash board Essentials and Metrics monitoring.

Books/Suggested References:

1. Jeeva Jose, Introduction to Machine Learning, Khanna Book Publishing 2020.
2. Rajiv Chopra, Machine Learning, Khanna Book Publishing 2021
3. Optimization for Machine Learning, Suvrit Sra, Sebastian Nowozin and Stephen J. Wright, MIT Press, 2011.
4. Optimization in Machine Learning and Applications, Suresh Chandra Satapathy, Anand J. Kulkarni, Springer, 2019

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Open Elective AI 604(B) Natural Language Processing

COURSE OBJECTIVES: Students should develop a basic understanding in natural language processing methods and strategies and to evaluate the strengths and weaknesses of various Natural Language Processing (NLP) methods & technologies and gain an insight into the application areas of Natural language processing.

Course outcomes: After completing the course student should be able to:

1. Define different data models used in Information Retrieval using NLP.
2. Demonstrate current methods for statistical approaches to machine translation.
3. Apply syntactic parsing and semantic analysis on text.
4. Solve and implement real world problems using NLP.

Detailed Contents:

UNIT I: Introduction: Origins and challenges of NLP – Grammar-based LM, Statistical LM – Regular Expressions, Finite-State Automata – English Morphology, Transducers for lexicon and rules, Tokenization, Detecting and Correcting Spelling Errors, Minimum Edit Distance.

UNIT II: Word Level Analysis: Unsmoothed N-grams, Evaluating N-grams, Smoothing, Interpolation and Backoff – Word Classes, Part-of-Speech Tagging, Rule-based, Stochastic and Transformation-based tagging, Issues in PoS tagging – Hidden Markov and Maximum Entropy models, Viterbi algorithms and EM training.

UNIT III: Syntactic Analysis: Context-Free Grammars, Grammar rules for English, Treebanks, Normal Forms for grammar – Dependency Grammar – Syntactic Parsing, Ambiguity, Dynamic Programming parsing – Shallow parsing – Probabilistic CFG, Probabilistic CYK, Probabilistic Lexicalized CFGs – Feature structures, Unification of feature structures.

UNIT IV: Semantics and Pragmatics: Requirements for representation, First-Order Logic, Description Logics – Syntax-Driven Semantic analysis, Semantic attachments – Word Senses, Relations between Senses, Thematic Roles, selectional restrictions – Word Sense Disambiguation, WSD using Supervised, Dictionary & Thesaurus, Bootstrapping methods – Word Similarity using Thesaurus and Distributional methods. Compositional semantics.

UNIT V: Application of NLP: intelligent work processors: Machine translation, user interfaces, Man-Machine interfaces, natural language querying, tutoring and authoring systems, speech recognition, and commercial use of NLP.

Text Books:

1. Daniel Jurafsky, James H. Martin—Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech, Pearson Publication.
2. Steven Bird, Ewan Klein and Edward Loper,—Natural Language Processing with Python, O'Reilly Media.
3. Manning and Schütze "Foundations of Statistical Natural Language Processing", MIT Press.

ReferenceBooks:

1. BreckBaldwin,LanguageProcessingwithJavaandLingPipeCookbook,Atlantic Publisher.
2. Richard M Reese, Natural Language Processing with Java, OReilly Media.
3. Nitin Indurkha and Fred J. Damerau, Hand book of Natural Language Processing, Chapman and Hall/CRC Press.
4. Tanveer Siddiqui, U.S. Tiwary, Natural Language Processing and Information Retrieval,Oxford University Press.

Course out comes: After completion of this course, the students would be able to:

- CO1. Understand the need of designing Enterprise data warehouses and will be enabled to approach business problems analytically by identifying opportunities to derive business.
- CO2. Compare and contrast, various methods for storing & retrieving data from different data sources/repository.
- CO3. Ascertain the application of data mining in various areas and Preprocess the given data and visualize it for a given application or data exploration/mining task
- CO4. Apply supervised learning methods to given data sets such as classification and its various types.
- CO5. Apply Unsupervised learning methods to given data sets such as clustering and its various types.
- CO6 Apply Association rule Mining to various domains.

Detailed Contents

1. Data Warehousing: Introduction, Delivery Process, Data warehouse Architecture, Data Preprocessing: Data cleaning, Data Integration and transformation, Data reduction. Data warehouse Design: Data warehouse schema, Partitioning strategy Data warehouse Implementation, Data Marts, Meta Data, Example of a Multidimensional Data model. Introduction to Pattern Warehousing.
2. OLAP Systems: Basic concepts, OLAP queries, Types of OLAP servers, OLAP operations etc. Data Warehouse Hardware and Operational Design: Security, Backup And Recovery,
3. Introduction to Data & Data Mining : Data Types, Quality of data, Data Preprocessing, Similarity measures, Summary statistics, Data distributions, Basic data mining tasks, Data Mining V/s knowledge discovery in databases. Issues in Data mining. Introduction to Fuzzy sets and fuzzy logic.
4. Supervised Learning: Classification: Statistical-based algorithms, Distance-based algorithms, Decision tree-based algorithms, Neural network-based algorithms, Rule-based algorithms, Probabilistic Classifiers
5. Clustering & Association Rule mining : Hierarchical algorithms, Partitional algorithms, Clustering large databases – BIRCH, DBSCAN, CURE algorithms. Association rules : Parallel and distributed algorithms such as Apriori and FP growth algorithms.

Books Recommended:

Text Books:

1. Pang-ning Tan, Steinbach & Kumar, "Introduction to Data Mining", Pearson Edu, 2019.
2. Jaiwei Han, Micheline Kamber, "Data Mining : Concepts and Techniques", Morgan Kaufmann Publishers.

ReferenceBooks:

1. Margaret H. Dunham, "Data Mining: Introductory and Advanced topics ", Pearson Edu., 2009.
2. Anahory&Murray, "DataWarehousingintheRealWorld", PearsonEdu., 2009.